

MA 523: Numerical Analysis

Course Code	MA 523
Course Name	Numerical Analysis
L-T-P-C	3-1-0-4
Prerequisites	NA
Intended for	M.Sc./B. Tech./M.S./Ph.D.
Distribution	Core for M.Sc. in applied Mathematics and Elective for other discipline
Approval	10th Senate (Jun 2016)

Course Contents

- **Module I** [3 Lectures]
Computer arithmetic, Kind of errors in Numerical Procedures, Significant digits, Backward error analysis, Sensitivity and conditioning, Stability and accuracy, Evolution of polynomials.
- **Module II** [6 Lectures]
Nonlinear Equations: Bisection method, Secant method, Newton's method, Method of False-position, Secant method, Fixed point iterations, order of convergence, Newtons method for multiple roots, Newtons methods and fixed point method for the system of nonlinear equations.
- **Module III** [7 Lectures]
Existence and uniqueness of interpolating polynomial, Lagrange polynomials, Divided differences, Evenly spaced points, Error of interpolation, Piecewise interpolation, Cubic spline, Least-Square approximations.
- **Module IV** [11 Lectures]
System of linear equations: Gaussian elimination, Partial Pivoting, Pivoting and Scaling in Gaussian Elimination method, Singular matrices, Determinants and Matrix inversions, Tridiagonal systems, Norms, Condition numbers and errors in solutions; Iterative methods: Jacobi, Gauss-Seidel and SOR Methods, Power method, Inverse power method and QR methods for finding the eigenvalues and eigenvectors of matrices.
- **Numerical Differentiation and Integration** [5 Lectures]
Numerical differentiation, Newton-Cotes integration formulae, Composite rules, Error terms for Newton - Cotes formulae and composite rules, Methods of undetermined parameters, Gaussian quadrature.
- **Initial and Boundary Value Problem (IVP & BVP)** [10 Lectures]
Taylor series method, Forward Euler, Backward Euler and Modified Euler methods, Runge-Kutta methods; Multistep methods: Milne's method, Adams-Moulton method, System of equations and Higher order equations, Stiff equations, Finite difference methods and Shooting methods for the Boundary value problems.

Text Books

1. K. E. Atkinson, **An Introduction to Numerical Analysis**, 2nd Edition, John Wiley, 2008.
2. R. L. Burden and J. D. Faires, **Numerical Analysis**, 7th Edition, Thomson Learning, 2001.

References

1. M. T. Heath, **Scientific Computing: An Introductory Survey**, McGraw Hill, 2002.
2. Brian Bradie, **A friendly introduction to Numerical Analysis**, Pearson Education, 2007.